EFFAT UNIVERSITY
COLLEGE OF ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE

جامعة عفت
EFFAT UNIVERSITY

CS 3081 - Fall 2026

LAB 3: PROPOSITIONAL LOGIC & MODEL CHECKING

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Task 1.1

KNOWLEDGE BASE

The program prints True because rain is true in every model that satisfies the knowledge base (KB). Only one row satisfies the KB, and rain is true in that row. Therefore, the KB entails rain.

Rain	Hagrid	Dumbledore	Knowledge Base
True	True	True	False
True	True	False	False
True	False	True	True
True	False	False	False
False	True	True	False
False	True	False	False
False	False	True	False
False	False	False	False

CODE 1.1

harry.py

```python
from logic import *

rain = Symbol("rain")
hagrid = Symbol("hagrid")
dumbledore = Symbol("dumbledore")

knowledge = And(
    Implication(Not(rain), hagrid),
    Or(hagrid, dumbledore),
    Not(And(hagrid, dumbledore)),
    dumbledore
)

print(model_check(knowledge, rain))
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\reeem\OneDrive\Desktop\CS 3081\Labs\Lab2> py harry.py
True
```

Task 1.2

CLUE PUZZLE

THERE MUST BE EXACTLY ONE PERSON, ONE ROOM, AND ONE WEAPON.

- **Ms. Scarlet:** Must be true because there must be a person, and Mustard and Plum are false, so they are ruled out.
- **Library:** Must be true because there must be a room, and Ballroom and Kitchen are false, so they are ruled out..
- **Knife:** Weapons, and Revolver and Wrench are ruled out. The revolver is ruled out by the initial cards. The wrench is ruled out by the unknown card, which says not all of Scarlet, library and wrench are true. Since Scarlet and library are both true, the wrench must be false, that leaves the knife.

Output: MsScarlet, Library, and Knife.

CODE 1.2

clue.py

```python
import termcolor

from logic import *

mustard = Symbol("ColMustard")
plum = Symbol("ProfPlum")
scarlet = Symbol("MsScarlet")
characters = [mustard, plum, scarlet]

ballroom = Symbol("ballroom")
kitchen = Symbol("kitchen")
library = Symbol("library")
rooms = [ballroom, kitchen, library]

knife = Symbol("knife")
revolver = Symbol("revolver")
wrench = Symbol("wrench")
weapons = [knife, revolver, wrench]

symbols = characters + rooms + weapons


def check_knowledge(knowledge):
    for symbol in symbols:
        if model_check(knowledge, symbol):
            termcolor.cprint(f"{symbol}: YES", "green")
        elif not model_check(knowledge, Not(symbol)):
            print(f"{symbol}: MAYBE")


# There must be a person, room, and weapon.
knowledge = And(
    Or(mustard, plum, scarlet),
    Or(ballroom, kitchen, library),
    Or(knife, revolver, wrench)
)
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\reeem\OneDrive\Desktop\CS 3081\Labs\Lab2> py clue.py
MsScarlet: YES
library: YES
knife: YES
```

Task 2.1 - 2.2

TASK	2.1 REMOVING A FACT	2.2 REMOVING A CLUE
CHANGE	Remove the final fact, dumbledore.	Remove the clue MinervaGryffindor.
PREDICTION	The program will print False.	Multiple house assignments will be possible.
RESULT	The program printed False.	Eight valid assignments remain, and the program prints no house assignments.
EXPLANATION	Without this fact, the KB no longer entails rain. This means rain is not guaranteed by the remaining facts.	No person–house assignment is true in every valid model, so the KB cannot determine anyone's house uniquely.

CODE 2.1

harry.py

```python
from logic import *

rain = Symbol("rain")
hagrid = Symbol("hagrid")
dumbledore = Symbol("dumbledore")

knowledge = And(
    Implication(Not(rain), hagrid),
    Or(hagrid, dumbledore),
    Not(And(hagrid, dumbledore)),

)

print(model_check(knowledge, rain))

```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\reeem\OneDrive\Desktop\CS 3081\Labs\Lab2> py harry.py
False
```

Task 2.2

ADDING A NEW CLUE

New clue: knowledge.add(Symbol("MinervaRavenclaw"))

Reported output:
Gilderoy → Gryffindor
Pomona → Hufflepuff
Minerva → Ravenclaw
Horace → Slytherin

Explanation: The new clue places Minerva in Ravenclaw. Together with the remaining constraints, it determines a unique house assignment for everyone.

CODE 2.2

puzzle.py

```
22
23  # Only one house per person.
24  for person in people:
25      for h1 in houses:
26          for h2 in houses:
27              if h1 != h2:
28                  knowledge.add(
29                      Implication(Symbol(f"{person}{h1}"), Not(Symbol(f"{person}{h2}")))
30                  )
31
32  # Only one person per house.
33  for house in houses:
34      for p1 in people:
35          for p2 in people:
36              if p1 != p2:
37                  knowledge.add(
38                      Implication(Symbol(f"{p1}{house}"), Not(Symbol(f"{p2}{house}")))
39                  )
40
41  knowledge.add(
42      Or(Symbol("GilderoyGryffindor"), Symbol("GilderoyRavenclaw"))
43  )
44
45  knowledge.add(
46      Not(Symbol("PomonaSlytherin"))
47  )
48
49
50  for symbol in symbols:
51      if model_check(knowledge, symbol):
52          print(symbol)
53
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\reeem\OneDrive\Desktop\CS 3081\Labs\Lab2> py puzzle.py
GilderoyRavenclaw
PomonaHufflepuff
MinervaGryffindor
HoraceSlytherin
PS C:\Users\reeem\OneDrive\Desktop\CS 3081\Labs\Lab2> py puzzle.py
PS C:\Users\reeem\OneDrive\Desktop\CS 3081\Labs\Lab2>
```

puzzle.py

```
24      for person in people:
25          for h1 in houses:
26              for h2 in houses:
27                  if h1 != h2:
31
32  # Only one person per house.
33  for house in houses:
34      for p1 in people:
35          for p2 in people:
36              if p1 != p2:
37                  knowledge.add(
38                      Implication(Symbol(f"{p1}{house}"), Not(Symbol(f"{p2}{house}")))
39                  )
40
41  knowledge.add(
42      Or(Symbol("GilderoyGryffindor"), Symbol("GilderoyRavenclaw"))
43  )
44
45  knowledge.add(
46      Not(Symbol("PomonaSlytherin"))
47  )
48
49  knowledge.add(
50      Symbol("MinervaRavenclaw"))
51
52  for symbol in symbols:
53      if model_check(knowledge, symbol):
54          print(symbol)
55
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\reeem\OneDrive\Desktop\CS 3081\Labs\Lab2> py puzzle.py
GilderoyGryffindor
PomonaHufflepuff
MinervaRavenclaw
HoraceSlytherin
PS C:\Users\reeem\OneDrive\Desktop\CS 3081\Labs\Lab2>
```

Task 3

EXPLANATION

How we solved it

We used propositional logic and model checking. Each person is a knight (always truthful) or a knave (always lies). We encoded what each person says with Implication(AKnight, statement) and Implication(AKnave, Not(statement)). Then model_check() found who must be a knight or a knave.

PUZZLE 0

Statement: A says “I am both a knight and a knave.”

Answer: A is a knave.

Why: Nobody can be both, so A’s statement is false and A must be lying.

PUZZLE 1

Statement: A says “We are both knaves.” B says nothing.

Answer: A is a knave, B is a knight.

Why: If A were a knight, the statement would be true, a contradiction. So A lies, and B is not a knave.

PUZZLE 2

Statement: A says “We are the same kind.” B says “We are of different kinds.”

Answer: A is a knave, B is a knight.

Why: If A were a knight, B would be one too, and B’s “different kinds” would be false, a contradiction. So A is a knave, they differ, and B is a knight.

PUZZLE 3

Statements: A says either “I am a knight” or “I am a knave.” B says “A said ‘I am a knave’” and “C is a knave.” C says “A is a knight.”

Answer: A is a knight, B is a knave, C is a knight.

Why: Nobody can say “I am a knave”, so B’s claim about A is false, which makes B a knave. B lies, so C is a knight, and C says A is a knight.

CODE 3

```python
from logic import *

AKnight = Symbol("A is a Knight")
AKnave = Symbol("A is a Knave")
BKnight = Symbol("B is a Knight")
BKnave = Symbol("B is a Knave")
CKnight = Symbol("C is a Knight")
CKnave = Symbol("C is a Knave")

# Puzzle 0: A says "I am both a knight and a knave."
knowledge0 = And(
    Or(AKnight, AKnave),
    Not(And(AKnight, AKnave)),
    Implication(AKnight, And(AKnight, AKnave)),
    Implication(AKnave, Not(And(AKnight, AKnave)))
)

# Puzzle 1: A says "We are both knaves." B says nothing.
knowledge1 = And(
    Or(AKnight, AKnave), Not(And(AKnight, AKnave)),
    Or(BKnight, BKnave), Not(And(BKnight, BKnave)),
    Implication(AKnight, And(AKnave, BKnave)),
    Implication(AKnave, Not(And(AKnave, BKnave)))
)

# Puzzle 2: A says "We are the same kind." B says "We are of different kinds."
knowledge2 = And(
    Or(AKnight, AKnave), Not(And(AKnight, AKnave)),
    Or(BKnight, BKnave), Not(And(BKnight, BKnave)),
    Implication(AKnight, Or(And(AKnight, BKnight), And(AKnave, BKnave))),
    Implication(AKnave, Not(Or(And(AKnight, BKnight), And(AKnave, BKnave)))),
    Implication(BKnight, Or(And(AKnight, BKnave), And(AKnave, BKnight))),
    Implication(BKnave, Not(Or(And(AKnight, BKnave), And(AKnave, BKnight))))
)

# Puzzle 3
# A says either "I am a knight" or "I am a knave"
# B says "A said 'I am a knave'" and "C is a knave." C says "A is a knight."

knowledge3 = And(
    Or(AKnight, AKnave), Not(And(AKnight, AKnave)),
    Or(BKnight, BKnave), Not(And(BKnight, BKnave)),
    Or(CKnight, CKnave), Not(And(CKnight, CKnave)),

    # B says "A said 'I am a knave'"
    Biconditional(BKnight, Biconditional(AKnight, AKnave)),
    # B says "C is a knave"
    Biconditional(BKnight, CKnave),
    # C says "A is a knight"
    Biconditional(CKnight, AKnight)
)

# Print every fact that must be true in each puzzle

print("Puzzle 0")
for symbol in [AKnight, AKnave]:
    if model_check(knowledge0, symbol):
        print(symbol)

print("Puzzle 1")
for symbol in [AKnight, AKnave, BKnight, BKnave]:
    if model_check(knowledge1, symbol):
        print(symbol)

print("Puzzle 2")
for symbol in [AKnight, AKnave, BKnight, BKnave]:
    if model_check(knowledge2, symbol):
        print(symbol)

print("Puzzle 3")
for symbol in [AKnight, AKnave, BKnight, BKnave, CKnight, CKnave]:
    if model_check(knowledge3, symbol):
        print(symbol)
```

```
PS C:\Users\reeem\OneDrive\Desktop\CS 3081\Labs\Lab2> py knights.py
Puzzle 0
A is a Knave
Puzzle 1
A is a Knave
B is a Knight
Puzzle 2
A is a Knave
B is a Knight
Puzzle 3
A is a Knight
B is a Knave
C is a Knight
```

THANK YOU

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